Functionally Complete

【Definition】 A set of logical operators is called ***functionally complete*** if every compound proposition is logically equivalent to a compound proposition involving only this set of logical operators.

- $\wedge$, $\vee$, and $\neg$ form a functionally complete set of operators.
- $\wedge$ ($\vee$) and $\neg$ form a functionally complete set of operators.

Propositional Normal Forms

- Literal: ***p* or ¬*p***

【**Definition**】 Disjunctions (conjunctions) with one or more literals as disjuncts (conjuncts) are called ***disjunctive (conjunctive) clauses*****.** Disjunctive and conjunctive clauses are simply called clauses.

- Examples:
  - $q \vee r$ Disjunctive clause
  - $\neg q \wedge p$ Conjunctive clause
  - $p \wedge q \vee r$ Not a clause

Conjunctive Normal Form (CNF)

【**Definition**】A conjunction with one or more disjunctive clauses as its conjuncts is said to be in ***conjunctive normal form***.

$$(A_{11} \vee \ldots \vee A_{1n_1}) \wedge \ldots \wedge (A_{k1} \vee \ldots \vee A_{kn_K})$$

- Examples:
  - $p \wedge (q \vee r)$ √
  - $\neg q \wedge p$ √
  - $p \wedge ((p \wedge q) \vee r)$ x

Disjunctive Normal Form (DNF)

【**Definition**】A disjunction with one or more conjunctive clauses as its disjuncts is said to be in ***disjunctive normal form***.

$$(A_{11} \wedge \ldots \wedge A_{1n_1}) \vee \ldots \vee (A_{k1} \wedge \ldots \wedge A_{kn_K})$$

- Examples:
  - $(p \wedge q) \vee (p \wedge \neg q)$ √
  - $p \vee (q \wedge r)$ √
  - $\neg p \vee \mathrm{T}$ √
  - $\neg(p \wedge q) \vee r$ x

More Examples

- p — DNF & CNF
- $\neg p \vee q$ — DNF & CNF
- $\neg p \wedge q \wedge \neg r$ — DNF & CNF
- $\neg p \vee (q \wedge \neg r)$ — DNF
- $\neg p \wedge (q \vee \neg r) \wedge (\neg q \vee r)$ — CNF

How to Obtain Normal Forms

Use logical Equivalences

1) $p \rightarrow q \Leftrightarrow \neg p \vee q$

2) $p \leftrightarrow q \Leftrightarrow (\neg p \vee q) \wedge (p \vee \neg q)$

3) $p \leftrightarrow q \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$

4) $\neg\neg p \Leftrightarrow p$

5) $\neg(p_1 \wedge \ldots \wedge p_n) \Leftrightarrow \neg p_1 \vee \ldots \vee \neg p_n$

6) $\neg(p_1 \vee \ldots \vee p_n) \Leftrightarrow \neg p_1 \wedge \ldots \wedge \neg p_n$

7) $p \wedge (q_1 \vee \ldots \vee q_n) \Leftrightarrow (p \wedge q_1) \vee \ldots \vee (p \wedge q_n)$
$(q_1 \vee \ldots \vee q_n) \wedge p \Leftrightarrow (p \wedge q_1) \vee \ldots \vee (p \wedge q_n)$

8) $p \vee (q_1 \wedge \ldots \wedge q_n) \Leftrightarrow (p \vee q_1) \wedge \ldots \wedge (p \vee q_n)$
$(q_1 \wedge \ldots \wedge q_n) \vee p \Leftrightarrow (p \vee q_1) \wedge \ldots \wedge (p \vee q_n)$

How to Obtain Normal Forms

Use logical Equivalences

- By (1)–(3) we eliminate $\rightarrow$ and $\leftrightarrow$.
- By (4)–(6) we eliminate $\neg$, $\wedge$, $\vee$, from the scope of $\neg$ such that any $\neg$ has only a literal as its scope.
- By (7) we eliminate $\vee$ from the scope of $\wedge$.
- By (8) we eliminate $\wedge$ from the scope of $\vee$.

This method leads to obtaining the disjunctive or conjunctive normal forms.

How to Obtain Normal Forms

- Examples

1. Convert the following formula into a conjunctive normal form.

$$\neg((p \vee \neg q) \wedge \neg r)$$

Solution:

$$\begin{aligned}
&\neg((p \vee \neg q) \wedge \neg r) \\
&\quad \equiv \neg(p \vee \neg q) \vee \neg\neg r && \text{De Morgan's Law} \\
&\quad \equiv \neg(p \vee \neg q) \vee r && \text{Double negation} \\
&\quad \equiv (\neg p \wedge \neg\neg q) \vee r && \text{De Morgan's Law} \\
&\quad \equiv (\neg p \wedge q) \vee r && \text{Double negation} \\
&\quad \equiv (\neg p \vee r) \wedge (q \vee r) && \text{Distributive law}
\end{aligned}$$

How to Obtain Normal Forms

- Examples

2. Convert the following formula into conjunctive and disjunctive normal forms $\neg($

Remark: **Formulas (*) is in CNF and formulas (**) is in DNF.**

Solution:

$$\neg(p \vee q) \leftrightarrow (p \wedge q)$$

$$\equiv (\neg(p \vee q) \rightarrow (p \wedge q)) \wedge ((p \wedge q) \rightarrow \neg(p \vee q))$$

$$\equiv ((p \vee q) \vee (p \wedge q)) \wedge (\neg(p \wedge q) \vee \neg(p \vee q))$$

$$\equiv ((p \vee q \vee p) \wedge (p \vee q \vee q)) \wedge ((\neg p \vee \neg q) \vee (\neg p \wedge \neg q))$$

$$\equiv (p \vee q) \wedge (\neg p \vee \neg q \vee \neg p) \wedge (\neg p \vee \neg q \vee \neg q)$$

$$\equiv (p \vee q) \wedge (\neg p \vee \neg q)^{*}$$

$$\equiv ((p \vee q) \wedge \neg p) \vee ((p \vee q) \wedge \neg q)$$

$$\equiv (p \wedge \neg p) \vee (q \wedge \neg p) \vee (p \wedge \neg q) \vee (q \wedge \neg q)$$

$$\equiv (q \wedge \neg p) \vee (p \wedge \neg q)^{**}$$

Full Disjunctive Normal Form

【**Definition**】 A ***minterm*** is a conjunction of literals in which each variable is represented exactly once.

- Example:

  If a Boolean function has the variables p, q, r, then

  $p \wedge \neg q \wedge r$ is a minterm, but $p \wedge \neg q$ and $p \wedge \neg p \wedge r$ are not.

- Question:

  If a formula has n variables, how many minterms are there? $\mathbf{2}^{n}$

【**Definition**】 If a formula is expressed as a disjunction of minterms, it is said to be in ***full disjunctive normal form***.

- Example:

$$(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$$

Transforming to Full Disjunctive Normal Form

1. Obtain disjunctive normal form,
2. Make use of negation laws and distributive laws to obtain full disjunctive normal form.

- Example

  Convert the following formula into full disjunctive normal form.

$$(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$$

Solution:

$$(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$$

$$\equiv (p \wedge q \wedge (r \vee \neg r)) \vee (\neg p \wedge (q \vee \neg q) \wedge r) \vee ((p \vee \neg p) \wedge q \wedge r)$$

$$\equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r) \vee (p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r)$$

$$\equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$$

Full Disjunctive Normal Form from Truth Table

$$f = (p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$$

p	*q*	*r*	*f*
T	T	T	T
T	T	F	T
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	T
F	F	F	F

$$f \equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$$

Full Disjunctive Normal Form from Truth Table

p	q	$\neg p \wedge \neg q$	$\neg p \wedge q$	$p \wedge \neg q$	$p \wedge q$
T	T	F	F	F	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	F	F	F

(1) Each minterm is true for exactly one assignment.

(2) If A and B are two distinct minterms, then $A \wedge B \Leftrightarrow$F.

(3) A disjunction of minterms is true only if at least one of its constituent minterms is true.

Full Disjunctive Normal Form from Truth Table

- Example

Find the full disjunctive normal form for *f* given by the table.

p	*q*	*r*	*f*
T	T	T	T
T	T	F	F
T	F	T	T
T	F	F	F
F	T	T	F
F	T	F	F
F	F	T	T
F	F	F	F

$$f \equiv (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$$

Full Disjunctive Normal Form

- Questions:

Using the full disjunctive normal form,

1. how to determine whether two statements are logical equivalent?
2. how to determine whether a given statement is tautology, contradiction or contingency?
3. how to find the assignments for which a given statement is true?

Prenex Normal Form

【**Definition**】A statement is in ***prenex normal form*** iff it is of the form $Q_1 x_1 Q_2 x_2 \cdots Q_n x_n B$, where $Q_i (i = 1, \cdots, n)$ is $\forall$ *or* $\exists$ and the predicate B is quantifier free.

- A formula with no quantifiers is regarded as a trivial case of a prenex normal form.

- Examples:
  - $\forall x P(x) \vee \forall x Q(x)$ X
  - $\forall x \forall y \neg (P(x) \rightarrow Q(y))$ √
  - $\forall x \exists y R(x,y)$ √
  - $\neg \forall x R(x)$ X

Transforming to Prenex Normal Form

- Any expression can be converted into a prenex normal form.

1. Eliminate all occurrences of $\rightarrow$ and $\leftrightarrow$ from the formula in question;
2. Move all negations inward such that, in the end, negations only appear as part of literals;
3. Rename the variables (when necessary);
4. The prenex normal form can now be obtained by moving all quantifiers to the front of the formula.

Transforming to Prenex Normal Form

- Step 1:
  - $A \rightarrow B \Leftrightarrow \neg A \vee B$
  - $A \leftrightarrow B \Leftrightarrow (\neg A \vee B) \wedge (A \vee \neg B)$
  - $A \leftrightarrow B \Leftrightarrow (A \wedge B) \vee (\neg A \wedge \neg B)$

- Step 2:
  - $\neg\neg A \Leftrightarrow A$
  - $\neg\exists x A(x) \Leftrightarrow \forall x \neg A(x)$
  - $\neg\forall x A(x) \Leftrightarrow \exists x \neg A(x)$

Transforming to Prenex Normal Form

- Step 3: Rename all variables in the statement
- Example:

$$\forall x(P(x)\rightarrow Q(x)) \wedge \exists x Q(x) \wedge \exists z P(z) \wedge \exists z(Q(z)\rightarrow R(z))$$

Solution:

Use y for x in $\forall x$, u for x in $\exists x Q(x)$, and w for z in $\exists z P(z)$:

$$\forall y(P(y)\rightarrow Q(y)) \wedge \exists u Q(u) \wedge \exists w P(w) \wedge \exists z(Q(z)\rightarrow R(z))$$

Transforming to Prenex Normal Form

- Step 4:
  - $A \wedge \exists x B(x) \Leftrightarrow \exists x\,(A \wedge B(x))$
  - $A \wedge \forall x B(x) \Leftrightarrow \forall x\,(A \wedge B(x))$
  - $A \vee \exists x B(x) \Leftrightarrow \exists x\,(A \vee B(x))$
  - $A \vee \forall x B(x) \Leftrightarrow \forall x\,(A \vee B(x))$
  - $\forall x \forall y\, A(x,y) \Leftrightarrow \forall y \forall x\, A(x,y)$
  - $\exists x \exists y\, A(x,y) \Leftrightarrow \exists y \exists x\, A(x,y)$
  - $Q_1 x A(x) \wedge Q_2 y B(y) \Leftrightarrow Q_1 x Q_2 y (A(x) \wedge B(y))$
  - $Q_1 x A(x) \vee Q_2 y B(y) \Leftrightarrow Q_1 x Q_2 y (A(x) \vee B(y))$

Transforming to Prenex Normal Form

- Example: Convert the following formulas into prenex normal form.

$$\forall x((\exists y R(x,y) \wedge \forall y \neg S(x,y)) \rightarrow \neg(\exists y M(x,y) \wedge P))$$

Solution:

$$\forall x((\exists y R(x,y) \wedge \forall y \neg S(x,y)) \rightarrow \neg(\exists y M(x,y) \wedge P))$$

$$\Leftrightarrow \forall x(\neg(\exists y R(x,y) \wedge \forall y \neg S(x,y)) \vee \neg(\exists y M(x,y) \wedge P))$$

$$\Leftrightarrow \forall x((\neg\exists y R(x,y) \vee \neg\forall y \neg S(x,y)) \vee (\neg\exists y M(x,y) \vee \neg P))$$

$$\Leftrightarrow \forall x((\forall y \neg R(x,y) \vee \exists y S(x,y)) \vee (\forall y \neg M(x,y) \vee \neg P))$$

$$\Leftrightarrow \forall x(\forall y \neg R(x,y) \vee \exists z S(x,z) \vee \forall u \neg M(x,u) \vee \neg P)$$

$$\Leftrightarrow \forall x \forall y \exists z \forall u(\neg R(x,y) \vee S(x,z) \vee \neg M(x,u) \vee \neg P)$$

Prenex CNF and DNF

- Step1: Prenex normal form
- Step 2: Prenex DNF or CNF

HW (Due on Mar. 14)

1. Give the simplest DNF and CNF of the following statements:

   1) $((p \to q) \leftrightarrow (\neg q \to \neg p)) \wedge r$

   2) $p \vee (\neg p \vee (q \wedge \neg q))$

   3) $(p \wedge (q \wedge s)) \vee (\neg p \wedge (q \wedge s))$

2. Give the full DNF of the following statements:

   1) $(\neg p \vee \neg q) \to (p \leftrightarrow \neg q)$

   2) $(\neg r \wedge (q \to p)) \to (p \to (q \vee r))$

   3) $(p \to (q \wedge r)) \wedge (\neg p \to (\neg q \wedge \neg r))$

HW (Due on Mar. 14)

3. Give the prenex normal forms of the following statements:

   1) $\forall x(P(x) \rightarrow \exists y Q(x,y))$

   2) $\forall x \forall y((\exists z P(x,y,z) \wedge \exists u Q(x,u)) \rightarrow \exists v Q(y,v))$

4. Give the prenex DNF and CNF of the following statements:

   1) $(\exists x P(x) \vee \exists x Q(x)) \rightarrow \exists x\, (P(x) \vee Q(x))$

   2) $\forall x \forall\, y\, (P(x) \rightarrow Q(x,y)) \rightarrow \exists y(\mathrm{P}(y) \wedge \exists z Q(y,z))$